ARMORER'S BENCH TECHNICAL MANUAL

Winchester Model 70

Winchester Repeating Arms • 'The Rifleman's Rifle' (Pre-64 & Classic Controlled Round Feed .30-06 / .270)

Manufacturer:	Winchester Repeating Arms	**Category:**	Rifles & Carbines
Action Type:	Controlled Round Feed (CRF) Mauser-Pattern Bolt Action	**Era / Service:**	American Hunting Benchmark (1936–Present)
Calibers:	.30-06 Springfield, .270 Winchester, .308 Win, .375 H&H; Magnum	**Covered Models:**	Pre-64 Model 70 (Collector Gold Standard), Classic Super Grade, Extreme Weather SS, Featherweight

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Mauser Non-Rotating Claw Extractor Tuning • Coned Breech Headspacing • 3-Position Wing Safety Shroud Timing • Pre-64 Trigger Stoning • Glass Bedding Flat-Bottom Receiver

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
CONTROLLED ROUND FEED	Massive non-rotating spring-steel claw extractor captures cartridge rim as it rises from magazine feed lips, guiding it into chamber
3 POSITION SAFETY	Position 1 (Rear): Safe & bolt locked; Position 2 (Mid): Safe & bolt can be opened to unload; Position 3 (Forward): Fire
CONED BREECH	Breech face of barrel is chamfered at 45° to match bolt head cone, ensuring bullet tip glides into chamber even if rifle is upside down
RECEIVER BOTTOM	Integral flat-bottom receiver with square recoil lug, ideal for epoxy pillar bedding

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Claw Extractor Won't Snap Over Single-Loaded Round Shooter drops single cartridge directly into chamber and tries to force bolt closed; bolt jams stubbornly	• Model 70 is pure Controlled Round Feed; extractor claw is designed to capture rim from beneath, not snap over. • Forcing bolt closed against rim risks chipping claw face.	**Immediate Action: Press round into magazine well and feed forward from magazine. Gunsmith Action: Bevel the front entry face of the claw extractor at 45° with fine stone to allow emergency push-feed snap-over without breaking.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Winchester Model 70 Pre-64 Action & 3-Position Safety	Mark Novak / Anvil Gunsmithing	Masterclass on the Model 70 action: coned breech geometry, claw extractor tension, timing the 3-position safety sleeve, and trigger stoning.